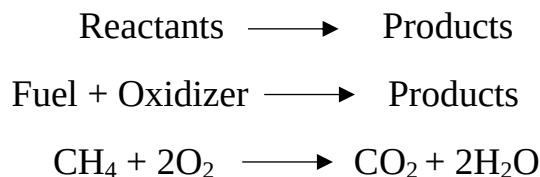


Thermodynamics of Combustion

Principle of Combustion: - A chemical reaction during which a fuel is oxidized and a large quantity of energy is released is called combustion. Complete combustion of a fuel is possible only in the presence of an adequate supply of oxidizer (oxygen, most often used oxidizer).

Combustion Process: - In the combustion process the reactants disappear and products appear as the reaction proceeds. Even though the masses of the fuel and oxidant do not remain constant in the combustion reaction, the masses of the constituent elements remain constant. Suppose methane CH_4 undergoes complete combustion in the presence of oxygen forming CO_2 and H_2O the combustion reaction can be represented as



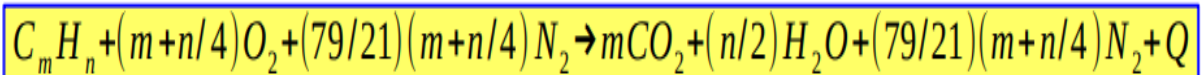
That is one mole of CH_4 reacts with 2 moles of O_2 yielding 1 mole of CO_2 and 2 moles of H_2O .

- If reaction produces heat $\rightarrow$ “exothermic”
- If reaction consumes heat $\rightarrow$ “endothermic”
- By definition, combustion is an exothermic process
- Chemical energy converted to thermal energy to do guided work
- Thermal energy is random

Stoichiometric or Theoretical Combustion Reaction

- The minimum amount of air that supplies sufficient oxygen for the complete combustion of all the carbon, hydrogen, and sulfur present in the fuel is called the theoretical amount of air.
- For complete combustion with the theoretical amount of air, the products consist of carbon dioxide, water, sulfur dioxide, the nitrogen accompanying the oxygen in the air, and any nitrogen contained in the fuel. No free oxygen appears in the products.

- The ideal combustion process during which a fuel is burned completely with theoretical air is called the stoichiometric or theoretical combustion of that fuel.
- In most combustion calculations dry air is assumed as a mixture of 79% (vol) N₂ and 21% (vol) O₂ or 3.764 moles of N₂ per mole O₂
- Molecular weight of pure air is 28.97, as it also contains small amounts of argon, carbon dioxide, hydrogen etc.



- Under real engine conditions, CO₂, CO, H₂, O₂, H, O, NO are also produced as a result of the dissociation reactions. Sulphur in the fuel produces SO₂ etc.

Complete and incomplete combustion

A combustion process is complete if all the carbon in the fuel burns to CO₂, all the hydrogen burns to H₂O, and all the sulfur (if any) burns to SO₂. That is, all the combustible components of a fuel are burned to completion during a complete combustion process.

Conversely, the combustion process is incomplete if the combustion products contain any unburned fuel or components such as C, H₂, CO, or OH.

Air–Fuel Ratio: - Two parameters that are frequently used to quantify the amounts of fuel and air in a particular combustion process are the air–fuel ratio and its reciprocal, the fuel–air ratio.

The air–fuel ratio is simply the ratio of the amount of air in a reaction to the amount of fuel. The ratio can be written on a molar basis (moles of air divided by moles of fuel) or on a mass basis (mass of air divided by mass of fuel).

Conversion between these values is accomplished using the molecular weights of the air, M_{air}, and fuel, M_{fuel},

$$\begin{aligned} \frac{\text{mass of air}}{\text{mass of fuel}} &= \frac{\text{moles of air} \times M_{\text{air}}}{\text{moles of fuel} \times M_{\text{fuel}}} \\ &= \frac{\text{moles of air}}{\text{moles of fuel}} \left(\frac{M_{\text{air}}}{M_{\text{fuel}}} \right) \end{aligned}$$

$$AF = \overline{AF} \left(\frac{M_{\text{air}}}{M_{\text{fuel}}} \right)$$

Where $\overline{AF}$ is the air–fuel ratio on a molar basis and AF is the ratio on a mass basis. For the combustion calculations the molecular weight of air is taken as 28.97.

Equivalence ratio equals the actual fuel–air ratio divided by the theoretical fuel–air ratio. Subscript s is used to indicate the ratio for 100% theoretical air, also called a stoichiometric mixture

$$\Phi = FA/FA_s = AF_s/AF$$

Equivalence ratio is also defines as the ratios of the stoichiometric air–fuel ratio and the actual air–fuel ratio, the molecular masses cancel out and they are the same whether a mass basis or a mole basis is used.

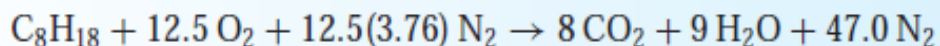
The reactants are said to form a lean mixture when the equivalence ratio is less than unity. When the ratio is greater than unity, the reactants are said to form a rich mixture.

Example. 1

Calculate the theoretical air–fuel ratio for the combustion of octane, C_8H_{18} .

Solution

The combustion equation is



The air–fuel ratio on a mole basis is

$$AF = \frac{12.5 + 47.0}{1} = 59.5 \text{ kmol air/kmol fuel}$$

The theoretical air–fuel ratio on a mass basis is found by introducing the molecular mass of the air and fuel.

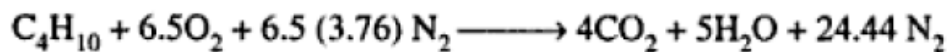
$$AF = \frac{59.5(28.97)}{114.2} = 15.0 \text{ kg air/kg fuel}$$

Example. 2

Calculate the stoichiometric air–fuel ratio on mass basis and on molar basis for the combustion of butane C_4H_{10} .

Solution :

The complete combustion reaction for butane in presence of air can be written as



Therefore, for the complete combustion of one mole of butane, 6.5 moles of O_2 and 24.44 moles of N_2 (that is $6.5 + 24.44 = 30.94$ moles of air) is to be supplied. Then

$$\text{Stoichiometric } A/F = \frac{30.94}{1} = 30.94 \text{ mol air / mol fuel}$$

Molar mass of air = $28.97 \times 10^{-3} \text{ kg / mol}$ and

Molar mass of butane C_4H_{10} = $58 \times 10^{-3} \text{ kg / mol}$.

Therefore, stoichiometric air fuel ratio on mass basis is given by

$$\text{Stoichiometric } A/F = \frac{30.94 \times 28.97 \times 10^{-3}}{58 \times 10^{-3}} = 15.45 \frac{\text{kg air}}{\text{kg fuel}}$$

Example

Calculate the molar composition of the combustion products if butane is burned with theoretical air. Assume complete combustion.

Solution :

The combustion reaction is given by



Total moles of combustion products $n = \sum n_i$

or $n = 4 + 5 + 24.44 = 33.44$

$$\text{Therefore, } CO_2 = \frac{4}{33.44} \times 100 = 11.96\%$$

$$H_2O = \frac{5 \times 100}{33.44} = 14.95\%; \quad N_2 = \frac{24.44 \times 100}{33.44} = 73.09\%$$

Hence, the molar composition of combustion products is 11.96% CO_2 ; 14.95% H_2O and 73.09% N_2 .

Excess air: - In actual combustion processes, it is common practice to use more air than the stoichiometric amount to increase the chances of complete combustion or to control the temperature of the combustion chamber. The amount of air in excess of the stoichiometric amount is called excess air. The amount of excess air is usually expressed in terms of the stoichiometric air as percent excess air or percent theoretical air.

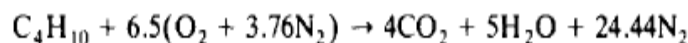
For example, 50 percent excess air is equivalent to 150 percent theoretical air, and 200 percent excess air is equivalent to 300 percent theoretical air. Of course, the stoichiometric air can be expressed as 0 percent excess air or 100 percent theoretical air.

$$\% \text{ theoretical air} = 100\% + \% \text{ excess air}$$

Amounts of air less than the stoichiometric amount are called deficiency of air and are often expressed as percent deficiency of air. For example, 90 percent theoretical air is equivalent to 10 percent deficiency of air.

EXAMPLE 12.1 Butane is burned with dry air at an air-fuel ratio of 20. Calculate (a) the percent excess air, (b) the volume percentage of CO_2 in the products, and (c) the dew-point temperature of the products.

The reaction equation for theoretical air is



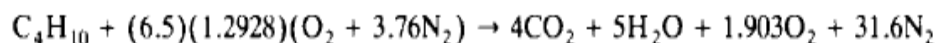
(a) The air-fuel ratio for theoretical air is

$$AF_{\text{th}} = \frac{m_{\text{air}}}{m_{\text{fuel}}} = \frac{(6.5)(4.76)(29)}{(1)(58)} = 15.47 \frac{\text{kg air}}{\text{kg fuel}}$$

This represents 100% theoretical air. The actual air-fuel ratio is 20. The excess air is then

$$\% \text{ excess air} = \left(\frac{AF_{\text{act}} - AF_{\text{th}}}{AF_{\text{th}}} \right) (100\%) = \left(\frac{20 - 15.47}{15.47} \right) (100\%) = 29.28\%$$

(b) The reaction equation with 129.28% theoretical air is



The volume percentage is obtained using the total moles in the products of combustion. For CO_2 we have

$$\% \text{ CO}_2 = \left(\frac{4}{42.5} \right) (100\%) = 9.41\%$$

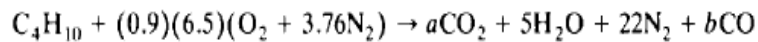
(c) To find the dew-point temperature of the products we need the partial pressure of the water vapor. It is found using the mole fraction to be

$$P_v = y_{\text{H}_2\text{O}} P_{\text{atm}} = \left(\frac{5}{42.5} \right) (100) = 11.76 \text{ kPa}$$

where we have assumed an atmospheric pressure of 100 kPa. Using Table C-2 we find the dew-point temperature to be $T_{\text{d.p.}} = 49^\circ\text{C}$.

EXAMPLE 12.2 Butane is burned with 90% theoretical air. Calculate the volume percentage of CO in the products and the air-fuel ratio. Assume no hydrocarbons in the products.

For incomplete combustion we add CO to the products of combustion. Using the reaction equation from Example 12.1,



With atomic balances on the carbon and oxygen we find:

$$\left. \begin{array}{l} \text{C:} \quad 4 = a + b \\ \text{O:} \quad 11.7 = 2a + 5 + b \end{array} \right\} \quad \therefore a = 2.7, b = 1.3$$

The volume percentage of CO is then

$$\% \text{ CO} = \left(\frac{1.3}{31} \right) (100\%) = 4.19\%$$

The air-fuel ratio is

$$AF = \frac{m_{\text{air}}}{m_{\text{fuel}}} = \frac{(0.9)(6.5)(4.76)(29)}{(1)(58)} = 13.92 \frac{\text{lbm air}}{\text{lbm fuel}}$$

WET AND DRY ANALYSIS OF PRODUCTS OF COMBUSTION

The analysis of the products of combustion can be given either on mass basis or volume basis. The products of combustion in hot condition have water in the vaporized form. The analysis made taking water vapour into account is called the wet analysis. If the analysis is made on the assumption that vapour has been condensed and removed it is called dry analysis. It must be emphasised here that merely cooling the products does not separate water vapour entirely. There will be some vapour still left at partial pressure corresponding to the temperature of products after cooling. The water vapour can however be removed entirely by using drying agent.

Analysis of Flue Gases by Orsat Apparatus

The volumetric analysis of dry products of combustion from a boiler or an engine is done by means of Orsat apparatus and by infrared gas analyser.

The most common method of analysing flue gases is by Orsat apparatus. It consists of a graduated burette tube connected to an aspirator-bottle by a rubber pipe. Aspirator-bottle is filled two third full water and sample can be drawn in or expelled out by lowering or raising aspirator. Three double absorption pipettes to absorb CO_2 , O_2 and CO respectively. The first pipette next to burette tube contains 40 per cent of KOH solution which would absorb twenty times its own volume of CO_2 . The second pipette contains pyrogalllic acid and KOH. This mixture may be made by mixing 5 gm of powdered pyrogalllic acid with 100 cc of 40 per cent KOH solution which would absorb twice its own volume of O_2 . The third pipette contains a solution of 5 gm of cuprous chloride in 100 cc of commercial HCl, which would absorb CO equal to its own volume.

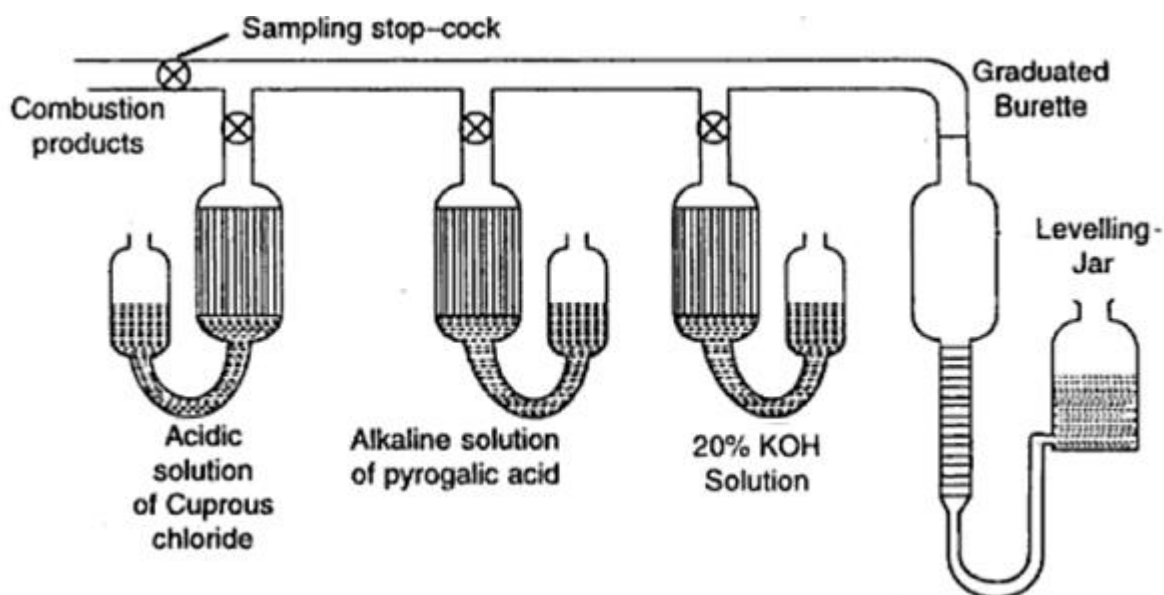


Figure: Schematic of Orsat Apparatus

Gas to be analysed may be collected directly into burette tube after removing air. After 100 cc of gas is taken into burette tube it is forced into first pipette by raising aspirator-bottle where CO_2 is absorbed. The liquid in the pipette is then levelled back by slowly lowering the aspirator-bottle and reading of burette is taken. The difference in the reading gives percentage of CO_2 by volume. The process is repeated such that the sample is forced through the other two pipettes also individually. The reduction in volume represents the partial volume of the component which has been absorbed. The Orsat analysis is carried out at temperature which is lower than the dew point temperature of the combustion products and hence the water vapor is not present in the combustion products analyzed by the Orsat apparatus. Thus the Orsat analysis gives the analysis on dry basis. The combustion products contain CO_2 , CO , O_2 and N_2 only and hence the unabsorbed gas is treated as N_2 .

Example

Methane (CH_4) is burned with atmospheric air. The analysis of the products on a dry basis is as follows:

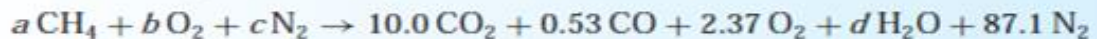
CO_2	10.00%
O_2	2.37
CO	0.53
N_2	<u>87.10</u>
	100.00%

Calculate the air–fuel ratio and the percent theoretical air and determine the combustion equation.

Solution

The solution consists of writing the combustion equation for 100 kmol of dry products, introducing letter coefficients for the unknown quantities, and then solving for them.

From the analysis of the products, the following equation can be written, keeping in mind that this analysis is on a dry basis.



A balance for each of the elements will enable us to solve for all the unknown coefficients:

Nitrogen balance: $c = 87.1$

Since all the nitrogen comes from the air

$$\frac{c}{b} = 3.76 \quad b = \frac{87.1}{3.76} = 23.16$$

Carbon balance: $a = 10.00 + 0.53 = 10.53$

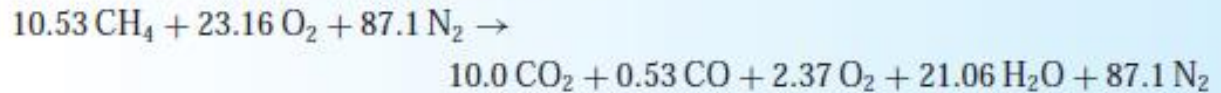
Hydrogen balance: $d = 2a = 21.06$

Oxygen balance: All the unknown coefficients have been solved for, and therefore the oxygen balance provides a check on the accuracy. Thus, b can also be determined by

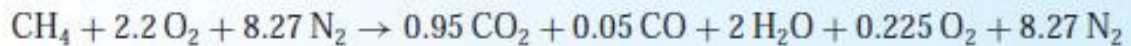
an oxygen balance

$$b = 10.00 + \frac{0.53}{2} + 2.37 + \frac{21.06}{2} = 23.16$$

Substituting these values for a , b , c , and d , we have



Dividing through by 10.53 yields the combustion equation per kmol of fuel.



The air–fuel ratio on a mole basis is

$$2.2 + 8.27 = 10.47 \text{ kmol air/kmol fuel}$$

The air–fuel ratio on a mass basis is found by introducing the molecular masses.

$$AF = \frac{10.47 \times 28.97}{16.0} = 18.97 \text{ kg air/kg fuel}$$

The theoretical air–fuel ratio is found by writing the combustion equation for theoretical air.



$$AF_{\text{theo}} = \frac{(2 + 7.52)28.97}{16.0} = 17.23 \text{ kg air/kg fuel}$$

The percent theoretical air is $\frac{18.97}{17.23} = 110\%$